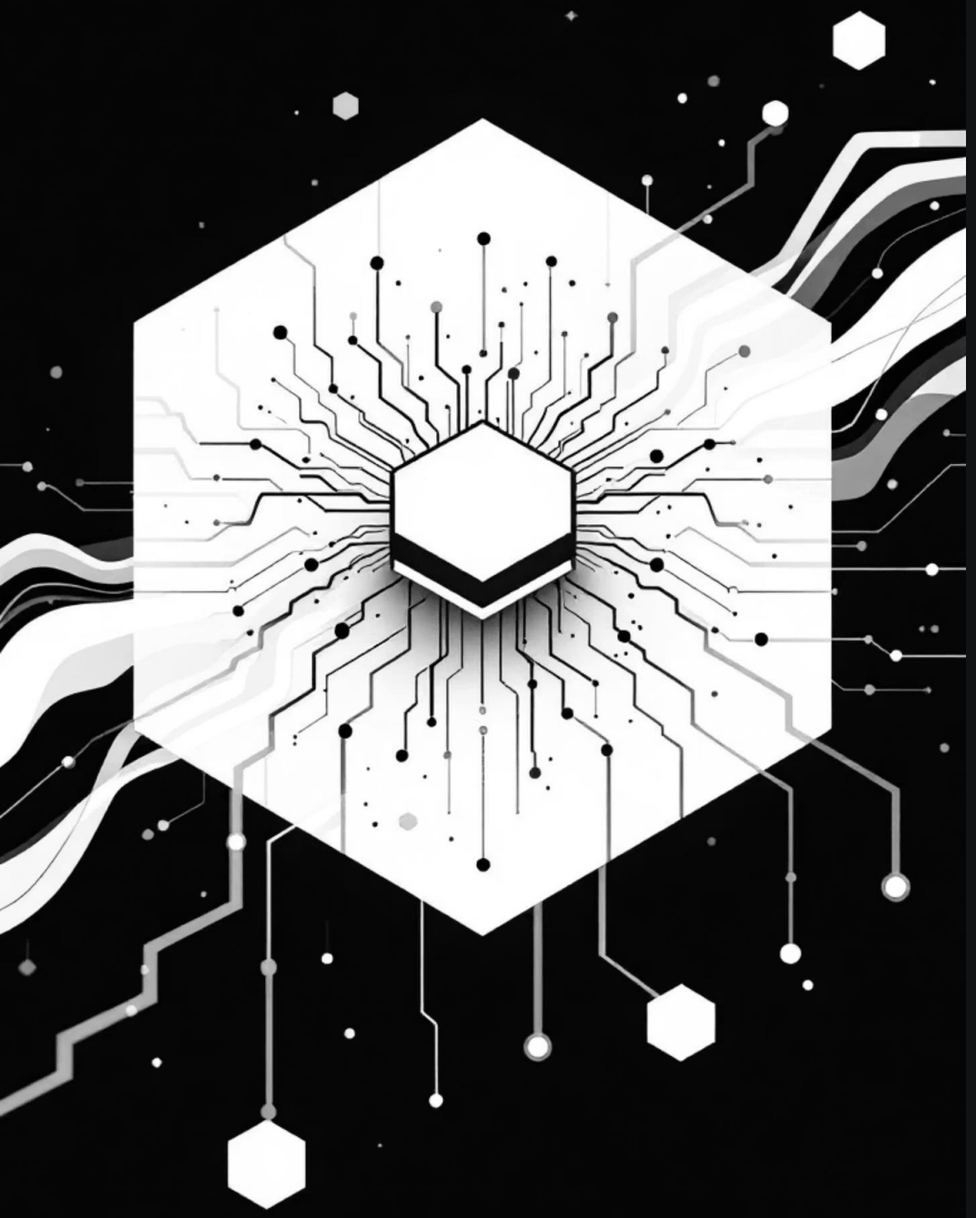


Introduction to Data Science

Turning Data into Decisions

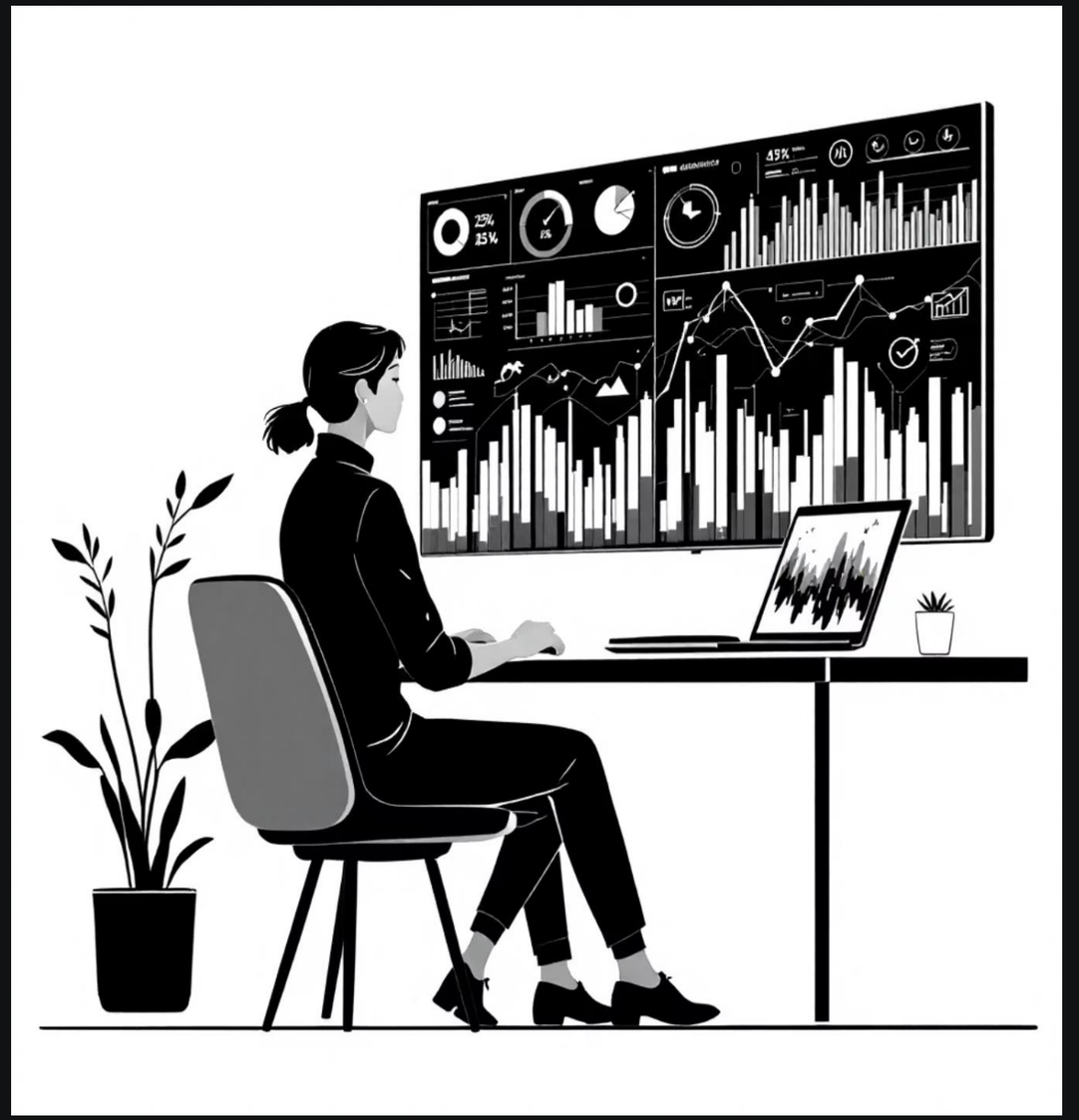
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Content

What You'll Learn Today

1. *What is Data Science*
2. *AI landscape overview (AI vs ML vs DS vs …)*
3. *Data Science Lifecycle*
4. *Data Science-related roles*



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Data Science

Data science is a multidisciplinary field that uses scientific methods, processes, algorithms, and systems to extract knowledge and insights from structured and unstructured data. It's a rapidly growing field with numerous applications in various industries.

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What is Data Science?

Data science is the practice of using scientific methods and algorithms to extract knowledge and insights from data. It involves collecting, cleaning, analyzing, and interpreting data to gain valuable information.

Data Collection

Gathering raw data from various sources, such as databases, APIs, and social media.

Data Analysis

Applying statistical methods and machine learning algorithms to identify patterns, trends, and relationships within the data.

Data Interpretation

Interpreting the findings and communicating them to stakeholders in a clear and concise manner.

Data Visualization

Presenting the data in a visually appealing and informative way to facilitate understanding.

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The AI Ecosystem: Understanding the Terminology

Artificial Intelligence (AI)

The broadest field encompassing any technique that enables machines to mimic human intelligence and perform tasks that typically require human cognition.

Examples: Chess-playing computers, voice assistants like Siri or Alexa, autonomous vehicles

Machine Learning (ML)

A subset of AI focused on algorithms that learn patterns from data and make predictions without being explicitly programmed for every scenario.

Examples: Spam email filters, product recommendation engines, fraud detection systems

Deep Learning (DL)

A specialized subset of ML using artificial neural networks with multiple layers. Excels at processing vast amounts of unstructured data like images, text, and audio.

Examples: Facial recognition on smartphones, ChatGPT, self-driving car vision systems

Data Science (DS)

The interdisciplinary field combining statistics, computer science, and domain expertise to extract knowledge and insights from both structured and unstructured data.

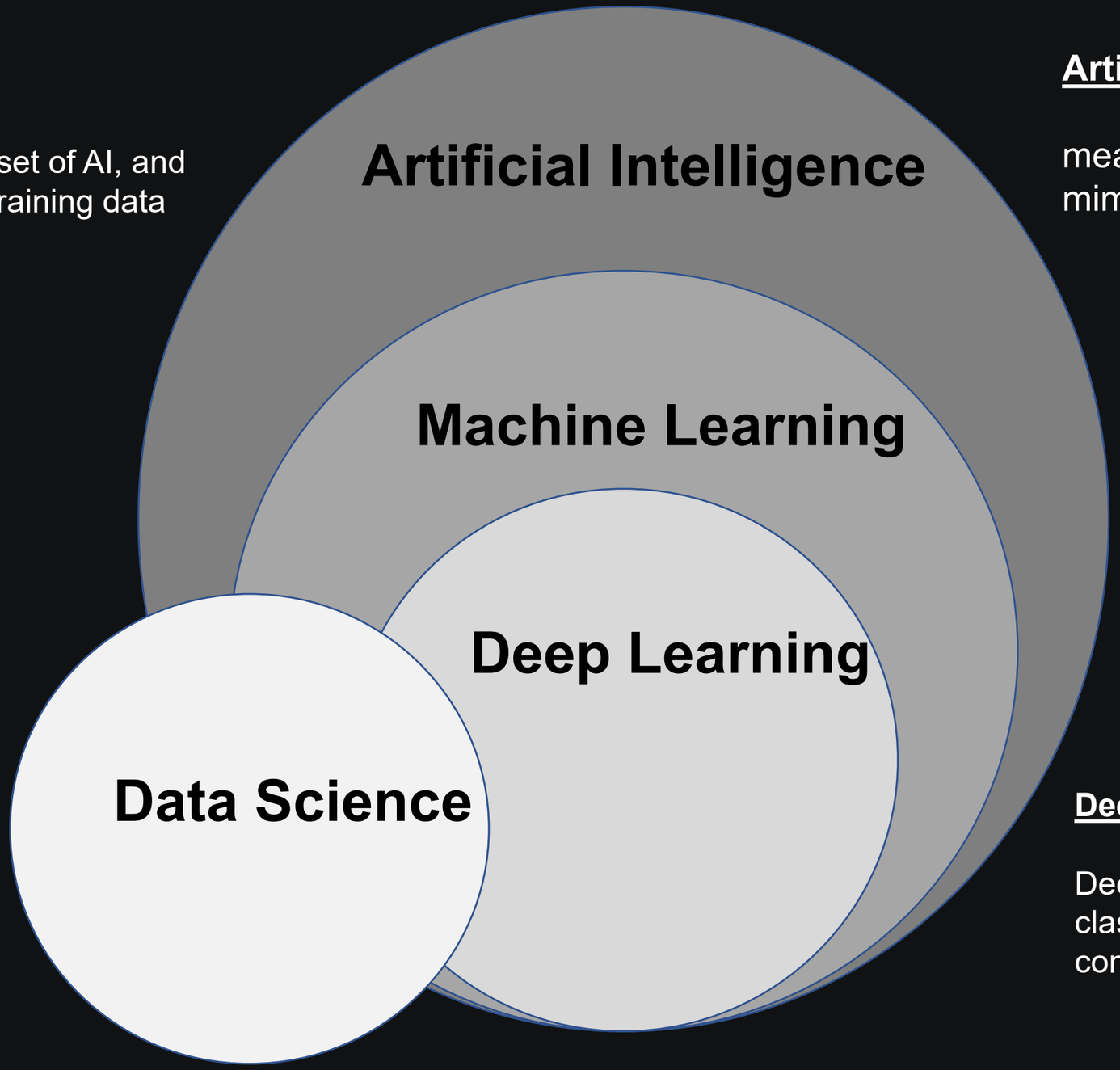
Key Distinction: Data Science is the entire process of problem-solving with data, which often—but not always—involves ML

Think of it this way:

If **AI** is the goal of building a "smart brain," **ML** is a key technique for achieving it, and **Data Science** is the comprehensive process of using data to solve real-world problems.

Machine Learning

Machine Learning is a subset of AI, and builds a model based on training data to make predictions.



Artificial Intelligence

means creating smart machines to mimic human behavior

Data Science

Data Science is a subset of AI. It is an area of statistics, scientific methods, etc. to extract meaning and insights from data.

Deep learning

Deep learning is a subset of ML, a class of ML algorithms to solve complex problems.

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The AI Ecosystem: Understanding the Terminology

Artificial Intelligence

The broadest field—making machines perform tasks requiring human intelligence

- Computer playing chess
- Voice assistants understanding commands
- Robots navigating environments

Machine Learning

A subset of AI where systems learn from data rather than explicit programming

- Spam filters improving over time
- Product recommendation engines
- Credit scoring models

Deep Learning

Advanced ML using neural networks, excelling with massive unstructured datasets

- Facial recognition systems
- Language models like ChatGPT
- Medical image analysis

Data Science

The interdisciplinary practice of extracting insights from data using statistics, programming, and domain expertise

- Customer behavior analysis
- Business forecasting
- A/B testing and experimentation

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Categorization Challenge: Test Your Understanding

1

Netflix Recommending Your Next Show

Answer: *Machine Learning & Data Science*

This system analyzes your viewing history, ratings, and behavior patterns (along with data from millions of other users) to predict what you'll enjoy watching next. It's a classic ML application embedded within a broader data science process.

2

Self-Driving Car Navigating a City

Answer: *AI (incorporating ML, DL, and DS)*

Autonomous vehicles represent the full AI spectrum: deep learning for visual recognition, machine learning for decision-making, and extensive data science for route optimization and safety protocols.

3

Analyzing Sales Data for Best Products

Answer: *Data Science (analytical focus)*

This scenario emphasizes descriptive and diagnostic analytics—looking at historical data to understand what happened and why. It may not require ML models at all, relying instead on statistical analysis and visualization.

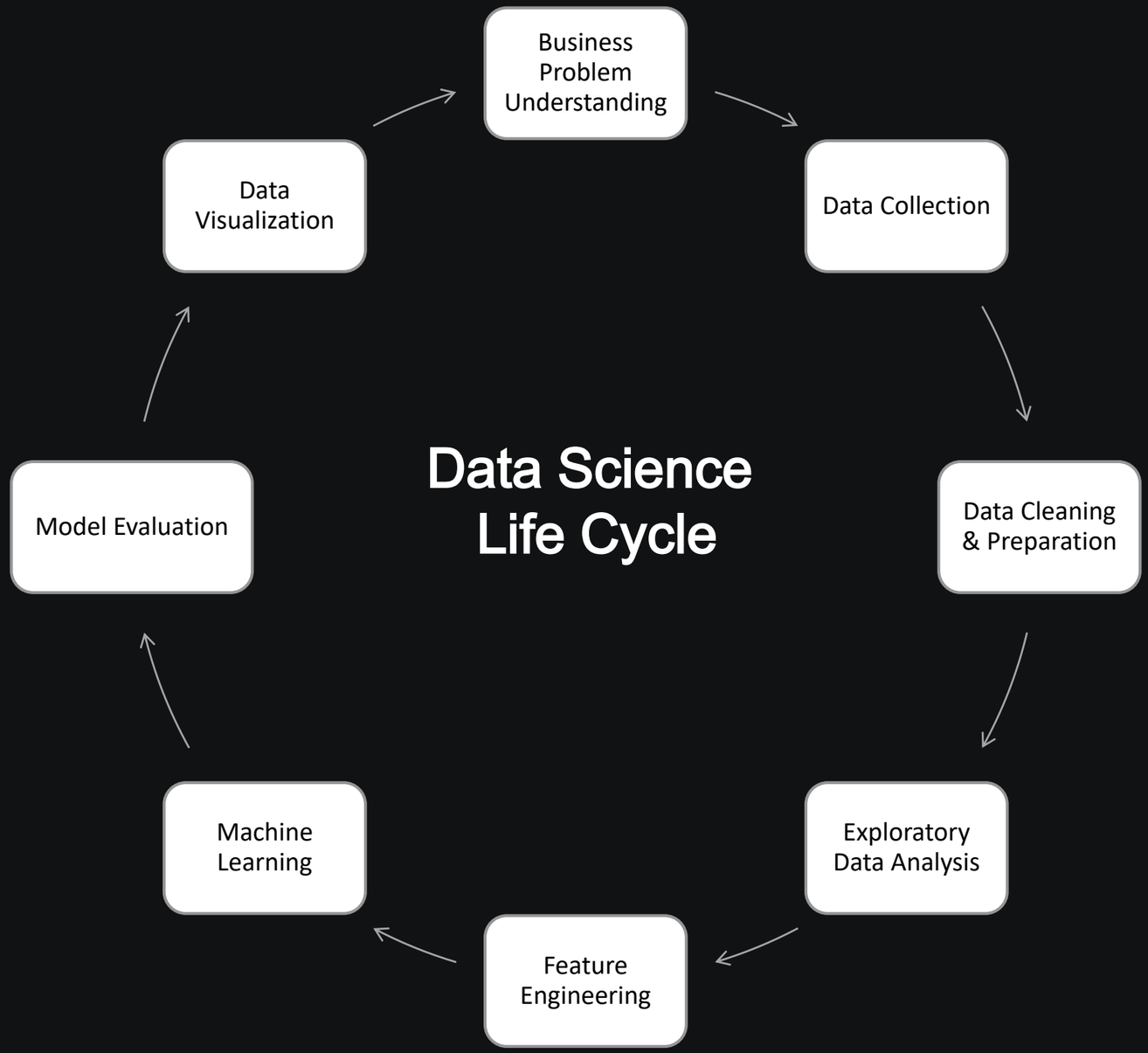
4

AI Generating Images from Text

Answer: *Deep Learning*

Systems like DALL-E or Midjourney use sophisticated deep neural networks trained on millions of image-text pairs to create entirely new, realistic images from natural language descriptions.

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The Data Science Lifecycle

Every successful data science project follows a structured process. We'll use a real-world scenario—predicting restaurant success—to illustrate each stage.



Business Understanding

Define the problem and success metrics. **Example:** Help investors identify high-potential restaurant locations.

Data Acquisition

Gather necessary data from multiple sources. **Example:** Demographics, foot traffic, competitor locations, rental prices, neighborhood reviews.

Data Preparation

Clean and format data. **Example:** Handle missing rental prices, standardize neighborhood names, merge datasets.

EDA & Modeling

Explore patterns and build predictive models. **Example:** Visualize success factors, train classification model for success/failure prediction.

Evaluation

Assess if the model meets business needs. **Example:** Test accuracy, precision, and real-world applicability for investment decisions.

Deployment

Put the model to work in production. **Example:** Create a web tool where investors input an address and receive a success score.

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Data Acquisition, Quality

Types of Data Sources

- **Internal Sources**

  Data your organization already owns and maintains within its systems.
  - Company databases (customer records, transactions)
  - User activity logs and clickstream data
  - CRM and ERP systems
  - Operational metrics and sensor data

  Advantages: Direct access, highly relevant, proprietary competitive advantage

- **External Sources**

  Data from outside your organization, ranging from free public datasets to premium commercial data.

  - Government open data portals (Census, weather, economic indicators)
  - Public datasets (Kaggle, UCI ML Repository, data.gov)
  - APIs (Twitter, Google Maps, financial data)
  - Commercial data providers
  - Web scraping (when legal and ethical)

  Considerations: Licensing, cost, quality verification, access limitations

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Data Acquisition, Quality

Standard Data/ML Dataset

1) IRIS
2) California house prediction
3) Wine bottle prediction
4) Car (Ammia...?)

Try dataset in Kaggle

What are the challenges in acquiring data?

1) Too many data
2) Biased
3) Reliable/consistency
4) Data is messy
5) Access

Where/how to get data ?

1. data.gov.my
2. Web scraping
3. Surveys
4. Statista
5. Kaggle
6. IEEE dataset
7. datahub.io
8. Shodan

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Data Collection and Preprocessing

Data collection involves gathering raw data from various sources, such as databases, APIs, and social media.

1 **Data Sources**

Identifying and selecting relevant data sources.

2 **Data Extraction**

Retrieving data from the chosen sources.

3 **Data Cleaning**

Addressing missing values, inconsistencies, and outliers.

4 **Data Transformation**

Converting data into a suitable format for analysis.

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Exploratory Data Analysis

Exploratory data analysis (EDA) involves analyzing and visualizing data to understand its characteristics, identify patterns, and gain insights.

Descriptive Statistics

Calculating measures of central tendency and dispersion.

Data Visualization

Creating graphs and charts to represent the data visually.

Hypothesis Testing

Evaluating relationships between variables and testing hypotheses.

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Machine Learning Algorithms

Machine learning algorithms are used to build predictive models based on data patterns. They can be used for tasks such as classification, regression, and clustering.

Algorithm	Task
Linear Regression	Predicting a continuous target variable.
Logistic Regression	Classifying data into two or more categories.
Decision Trees	Building a tree-like model to make decisions.
Support Vector Machines	Finding the optimal hyperplane to separate data points.





Model Evaluation and Deployment

Model evaluation involves assessing the accuracy and performance of the trained models using metrics like accuracy, precision, and recall.

1

Model Training

Training the model on a dataset.

2

Model Evaluation

Assessing the model's performance.

3

Model Optimization

Fine-tuning the model to improve its accuracy.

4

Model Deployment

Implementing the model into real-world applications.

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The Data Team: Different Roles, Shared Mission



Data Engineer



Data Analyst



Data Scientist



Machine Learning Engineer

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The Data Team: Different Roles, Shared Mission

Building data-driven solutions requires a diverse team. Understanding these roles helps you identify where your interests and skills align in the data ecosystem.



Data Engineer

Builds and maintains data infrastructure—the pipelines that collect, store, transform, and deliver data

Key Tools: Spark, Airflow, SQL, Cloud services, ETL tools

Focus: How do we move and manage data reliably?

Typical Output: Data warehouses, ETL pipelines, real-time data streams



Data Analyst

Interprets existing data to answer specific business questions and creates dashboards that inform decisions

Key Tools: SQL, Excel, Tableau, Power BI

Focus: What happened and why?

Typical Output: Monthly sales reports, customer behavior dashboards, KPI tracking



Data Scientist

Builds predictive models using advanced statistics and machine learning to solve complex, forward-looking problems

Key Tools: Python, R, SQL, scikit-learn, TensorFlow

Focus: What will happen and how can we optimize?

Typical Output: Predictive models, A/B test analyses, customer segmentation studies



ML Engineer

Deploys, scales, and maintains machine learning models in production systems, ensuring reliability at scale

Key Tools: Docker, Kubernetes, Cloud platforms, MLOps tools

Focus: How do we make this work in production?

Typical Output: Production ML pipelines, model APIs, automated retraining systems

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Your Data Science Journey Begins

What We've Covered

- The distinctions between AI, ML, and data science
- Diverse roles within data teams and their responsibilities
- The complete data science lifecycle from problem to deployment
- Data sources, types, and acquisition strategies

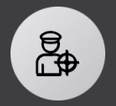
The Big Picture

Data science is a structured, systematic process for transforming raw data into decisions that drive real-world impact. It combines technical skills with critical thinking and domain expertise.



Remember

Every data project starts with understanding the business problem, not the technology



Collaborate

Successful data science requires diverse skills and perspectives working together



Stay Curious

The field evolves rapidly—continuous learning is part of the journey

Questions? The floor is open. Let's discuss anything from today's session or preview what's ahead in your data science education.

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Thank You

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